Approximation Algorithms and LP Relaxations for Vehicle Routing: Makespan, Latency, and Capacity Constraints

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Abstract

This document provides a comprehensive review of mathematical formulations, Linear Programming (LP) relaxations, and approximation algorithms for a class of vehicle routing problems. We consider routing n trucks from a single depot to visit K locations in a metric space. The primary objective is to minimize the arrival time of the last truck to return to the depot (makespan). We also explore extensions involving capacity constraints (where each location demands a certain amount of goods and each truck has a finite capacity) and alternative objectives, such as minimizing the weighted sum of arrival times (latency) at the locations, as well as bi-objective formulations combining makespan and latency.

1. Introduction

The problem of routing a fleet of vehicles to serve a set of geographically dispersed customers is fundamental in operations research and theoretical computer science. Given a set of K locations (or clients) $V = v_1, v_2, \dots, v_K$ and a depot r in a metric space with distance function $d(\cdot, \cdot)$, we are tasked with finding a set of tours for n trucks. All trucks start and end at the depot.

The core version of this problem aims to minimize the **makespan**, defined as the time when the last truck returns to the depot. This problem is known as the Min-Max Multiple Traveling Salesperson Problem (min-max mTSP). It is strictly NP-hard, as it generalizes the classical Traveling Salesperson Problem (TSP) when $n = 1$.

2. Minimizing Makespan: The Min-Max mTSP

2.1 Standard Mathematical Formulation

Let $x_{ij}^k \in 0, 1$ be a decision variable indicating whether truck k traverses the edge (i, j). Let $y_i^k \in 0, 1$ indicate if truck k visits location i. To minimize the maximum return time T, we formulate a Mixed Integer Linear Program (MILP):

Minimize T

Subject to:

2.2 The Configuration LP Relaxation

While the standard LP relaxation suffers from a large integrality gap, the **Configuration LP** provides a much tighter bound. Let T be a guessed optimal makespan. Let $\mathcal{C}_T$ be the set of all valid tours (starting and ending at the depot r) such that the total length of the tour is at most T. Let $x_C \geq 0$ be a fractional variable denoting whether configuration $C \in \mathcal{C}_T$ is selected. The LP determines if a fractional cover of size at most n exists for a given T:

Minimize $\sum_{C \in \mathcal{C}_T} x_C$

Subject to:

This LP is solved using **Column Generation**. The pricing problem asks: *Is there a tour of length at most T that collects enough dual weight?* This is the Resource-Constrained Shortest Path Problem, which admits FPTAS.

2.3 Approximation Algorithm: Tour Partitioning

Pseudo-code for the Tour Partitioning Algorithm: `text Algorithm: Tour-Partitioning-mTSP(V, r, n) 1. Find an approximately optimal TSP tour traversing all vertices in V ∪ {r}. (For example, use Christofides' Algorithm to yield a tour τ of length L). 2. Choose an arbitrary starting direction from the depot r along the tour τ. 3. Walk along the tour τ and cut it into exactly n contiguous segments (paths) P_1, P_2, ..., P_n, such that the length of each segment is exactly L/n. 4. For each truck k ∈ {1, ..., n}: a. Travel directly from depot r to the first node of path P_k. b. Traverse the path P_k, servicing all locations assigned to it. c. Travel`

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directly from the last node of P_k back to the depot r. 5. Return the n
constructed truck routes.
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2.4 Proof of the Approximation Ratio

Let $T^$ *be the optimal makespan. Lemma 1 (Lower Bounds): 1. Star Bound: $T^* \ge 2 \max_{v \in V} d(r, v). $Thus,$ d(r, v) \le \frac{T^*}{2}$. 2. TSP Bound: A single TSP tour covering all nodes has length $L_{OPT} \le n \cdot T^*$.

Theorem 1: *Tour Partitioning yields a 2.5-approximation constant factor for the min-max mTSP.* **Proof:** 1. Let τ be a TSP tour generated using Christofides ($\rho = 1.5$). Length $L \le 1.5 n T^. $2. The tour is cut into n segments, each of length$ L/n \le 1.5 T^* $. 3. By the triangle inequality, the route assigned to truck$ k $has length bounded by the distance to the path's start node$ u_k $, the path length, and the return from the path's end node$ v_k$: 4. Using the star bound, $d(r, u_k) \le \frac{T^*}{2} and d(v_k, r) \le \frac{T^*}{2}. $5. Therefore,$ \text{TourLength}(k) \le \frac{T^*}{2} + 1.5 T^* + \frac{T^*}{2} = 2.5 T^*

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3. Incorporating Capacity Constraints (CVRP)

When each location i demands g_i goods and trucks have maximum capacity G, the problem becomes the Capacitated VRP.

3.1 The Configuration LP

Let $\mathcal{C}*{G} $be the set of all tours where the total demand$ \sum*{v \in C} g_v \le G. Let d_C $be the metric length of tour$ C$. The Configuration LP for minimizing total distance (sum of makespans) is:

Minimize $\sum_{C \in \mathcal{C}_G} d_C x_C$

Subject to:

3.2 Approximation Algorithm: Iterated Tour Partitioning (ITP)

Haimovich and Rinnooy Kan (1985) introduced ITP for CVRP.

Pseudo-code: `text Algorithm: ITP-CVRP(V, r, G, g_i) 1. Find a ρ-approximate TSP tour τ starting at r. 2. Traverse τ, accumulating demand` `g_i. 3. Whenever the accumulated demand reaches exactly G, split the tour. (If a node's demand exceeds the remaining capacity, fractionally split its demand, or mathematically push the cut just before the node to strictly enforce G). 4. Connect the breakpoints back to the depot r to form independent truck tours. 5. Return the resulting tours.`

3.3 Proof of the Approximation Ratio

Let $L_{OPT}^{(CVRP)}$ be the optimal total routing cost. **Lemma 2 (Lower Bounds):** 1. **TSP Bound:** $L_{OPT}^{(CVRP)} \geq L_{OPT}^{(TSP)}$. 2. **Radial Bound:** Every node v must send its g_v demand to the depot, and each trip carries at most G. Thus:

Theorem 2: *ITP provides a 2.5-approximation constant factor for the min-sum CVRP.* **Proof:** 1. The length of the TSP tour τ is bounded by $L \leq \rho L_{OPT}^{(TSP)}$. 2. Slicing the tour into segments of capacity exactly G ensures that the extra distance incurred by traveling to and from the depot is exactly evaluated as a continuous integration of the radial distances. The total cost of the constructed tours is exactly the original TSP path length plus the sum of radial distances scaled by the cuts. 3. Specifically, the added radial cost can be bounded tightly by $\frac{2}{G} \sum_{v \in V} g_v d(r, v)$. 4. Combining the bounds: Using Christofides ($\rho = 1.5$), this yields a 2.5-approximation. \hfill ■

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4. Minimum Latency and Weighted Arrival Time (MLP)

The Minimum Latency Problem (MLP) shifts the objective to user-centric metrics: minimizing the sum of arrival times.

4.1 Time-Indexed Configuration LP

To model latency, variables must capture *time*. Let t be a discrete time step. Let y_v^t be a binary variable indicating if vertex v is visited by time t. Let $\mathcal{P}_t$ be the set of valid paths originating at r of length at most t. Let $x_P \geq 0$ denote selecting path P.

Minimize $\sum_{v \in V} \sum_{t=0}^{T_{\max}} (1 - y_v^t)$

Subject to: *(This elegantly captures that latency equals the area under the unvisited nodes curve).*

4.2 Approximation Algorithm: Geometric Scaling

For the single-vehicle MLP, Blum et al. (1994) propose a geometric scaling framework.

Pseudo-code: `text Algorithm: Geometric-Latency-Approximation(V, r) 1. Initialize an empty path P. Let D be the distance to the closest node. 2. For i = 1, 2, ..., max_iterations: a. Let length limit L_i = D * 2^i. b. Find a path rooted at r of length at most L_i that visits the MAXIMUM possible number of unvisited nodes. (This requires an approximation for the k-path problem / Orienteering). c. Append this path to P (traveling back to r between paths). 3. Return the concatenated path P.`

4.3 Proof of the Approximation Ratio

Theorem 3: *Geometric scaling yields a constant-factor approximation for MLP. Specifically, the best-known constant factor approximation for the single-vehicle case is 3.59 (Chaudhuri et al., 2003), and for the multi-vehicle case (k-TRP) it is 8.49 (Post and Swamy, 2015).* **Proof Sketch:** 1. Suppose a vertex v is optimally visited at time $t_v^. Intheoptimalsolution, thereisapathoflengtht_v^ coveringv. 2.Inouralgorithm, thereexistssomeiterationiwhereL_i \ge t_v^ \ge L_{i-1}. 3.Atiterationi, ourapproximatek −pathsubroutinewillfindapaththatis"dense"enoughtocoverv . 4.Theactualtimeouralgorithmreachesv$ *is bounded by the sum of lengths of all previous paths: 5. Since $L_i \le 2 t_v^, thearrivaltimeisO(t_v^*) . Summingthisconstantmultiplicativepenaltyoverallverticesyieldstheproven O(1)Unsupported use of \hfill\blacksquare$

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5. Bi-Objective Routing: Makespan and Latency

In practice, dispatchers need a solution minimizing the convex combination:

5.1 Bicriteria Mathematical Formulation

Instead of a strict convex combination, algorithms aim for a **Bicriteria (α, β)-Approximation**:

Minimize Total Latency

Subject to: *(Where T_{budget} is a user-defined hard constraint).*

5.2 Approximation Algorithm: Depth-First Tree Doubling

To achieve a bicriteria approximation, we leverage the properties of Minimum Spanning Trees (MST).

Pseudo-code: `text Algorithm: Bi-Objective-Routing(V, r, T_budget) 1. Solve the min-max mTSP using the Tour Partitioning algorithm to find a set of base trees that guarantee Makespan <= α * T_budget. 2. To optimize latency on these assigned trees, we do not perform arbitrary Eulerian tours. Instead, perform a Depth-First Search (DFS) traversal of the trees. 3. Order the children of each node in the DFS by the weight of their subtrees (visiting lighter subtrees first). 4. Return the resulting DFS-ordered tours.`

5.3 Proof of the Approximation Ratio

Theorem 4: *DFS-ordered tree doubling yields a Pareto-approximate frontier, specifically establishing a (2.5, 8.49)-bicriteria approximation.* **Proof Sketch:** 1. From Theorem 1, the makespan constraint is satisfied within a constant factor $\alpha = 2.5$. 2. To bound latency, consider a specific truck's tree. By visiting lighter subtrees first, we minimize the number of nodes waiting while the truck traverses deep, heavy branches. 3. Mathematically, it has been shown (e.g., by Goemans and Williamson) that an optimal DFS ordering of a tree guarantees the average arrival time of nodes on the tree is bounded by a constant factor β (established at 8.49) of the optimal latency constraint for that cluster. 4. Thus, the solution is simultaneously bounded by $\alpha \cdot \text{OPT}{*\text{text{makespan}}and*\beta \cdot *\text{text{OPT}}*{\text{latency}}Unsupported use of \hfill\blacksquare$

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6. Detailed Analysis of Related Work

The literature on vehicle routing approximations is vast, but several foundational papers establish the paradigms used to tackle makespan, capacity, and latency objectives.

1. Min-Max Tree Covers and Makespan: Even, Garg, Könemann, Ravi, and Sinha (2004) introduced a rigorous framework for finding min-max tree covers. By guessing the optimal tree weight and repeatedly solving Prize-Collecting Steiner Tree (PCST) instances, they provided a constant-factor approximation. *Relation to our problem:* This paper directly solves the min-max mTSP relaxation. To minimize the return time of the last truck, we can relax the problem to finding n bounded-weight trees, which are then Eulerian-doubled to form valid return tours.

2. Capacitated Routing and Iterated Tour Partitioning (ITP): Haimovich and Rinnooy Kan (1985) introduced the ITP framework. They demonstrated that for planar constraints or metric spaces, routing problems with capacity bounds can be solved by first constructing an optimal (or approximately optimal) single-vehicle TSP tour, and then optimally partitioning it into n segments where the sum of demands g_i in each segment does not exceed truck capacity G_j. *Relation to our problem:* When extending the LP to incorporate $g_i \leq G_j$, the ITP heuristic acts as the backbone for almost all practical approximation algorithms, providing a structural guarantee that breaking a giant tour incurs a bounded penalty on the makespan.

3. Min-Max Capacitated Vehicle Routing: Bompadre, Dror, and Orlin (2006) expanded upon ITP specifically for the min-max objective. Instead of just minimizing total distance, their work balances the length of the partitioned sub-tours to ensure no single truck takes on a disproportionate distance. *Relation to our problem:* Their work provides the exact approximation algorithms required when both capacities G_j and the min-max objective are present, bridging the gap between classical CVRP and mTSP.

4. Minimum Latency and the Traveling Repairman Problem: Blum, Chalasani, Coppersmith, Pulleyblank, Raghavan, and Sudan (1994) provided the first constant-factor approximation for the single-vehicle Minimum Latency Problem (MLP). They observed that the MLP is structurally different from TSP; optimal TSP tours can yield arbitrarily bad latency. *Relation to our problem:* This paper establishes why alternative objectives (like weighted arrival time) require entirely different LP relaxations (e.g., time-indexed or flow-based variables) rather than simple subtour elimination constraints.

5. Multi-Vehicle Latency (k-TRP): Fakcharoenphol, Harrelson, and Rao (2003) generalized the MLP to multiple vehicles (k-TRP). Their algorithm divides the time horizon into geometrically increasing intervals and uses k-MST approximations at each step. *Relation to our problem:* For our alternative objective (weighted arrival time with n trucks), their approximation bounds and geometric scaling techniques are directly applicable to bounding the maximum delay experienced by any location.

7. Conclusion

Routing n trucks to service K locations involves complex trade-offs between computational tractability and model fidelity. The LP relaxations for makespan (min-max mTSP) and capacity (CVRP) rely heavily on tree constraints and capacity cuts. Transitioning to latency (MLP) shifts the focus to flow and time-indexed constraints. Approximating a weighted combination of these objectives requires bicriteria techniques that balance the egalitarian nature of makespan against the utilitarian nature of latency. Future work could incorporate stochastic demands or time windows, further enriching the LP structures.

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